D9. Automated Evaluation Harness

Reproducible question-level and audit-level evaluation capability.

Field	Value
Project	Development of AI-Based Cyber Security Audit Assessment Against Benchmark
Scope	Objective 3; secure local/air-gapped RAG proof of concept
Report date	28 August 2026
Evidence rule	Implemented, tested and untested claims are labelled separately.

Reproducibility contract

Record	Captured by harness	Why a reviewer needs it
Input identity	Report ID, source hash and question ID	Connects a score to immutable inputs
Retrieval trace	Chunk IDs, scores and CRAG state	Explains what evidence reached the model
Generation trace	Model, prompt, seed, temperature and response	Supports controlled reruns
Scoring trace	Gold answer, correctness, grounding and audit roll-up	Separates local error from final decision
Release gate	Report/label/quality thresholds	Blocks overclaiming from small evidence

Harness behaviour demonstrated

The completed runs produce JSON and CSV question-level records, retrieval and classification metric files, faithfulness measures, audit scores, logs and exact configurations. This directly meets the requirement to trace every final audit score to its benchmark questions and evidence. [E4][E5][E10]

The production-readiness gate is deliberately stricter than the POC scorer. A report may score PASS against the selected benchmark while the evidence package is still BLOCKED from a production claim because the evaluation corpus is insufficient. [E5][E10]

Harness outputs

Artifact	Contents	Traceability value
question_level_results.json / .csv	Answer, retrieved evidence, scores, CRAG, latency, flags	Explains each benchmark decision [E5]
retrieval_metrics.json	Per-query and aggregate ranking measures	Separates retrieval from LLM quality [E5]
question_level_llm_metrics.json	Accuracy, macro-F1, per-class metrics, confusion matrix	Shows class imbalance and errors [E5]
sota_faithfulness_metrics.json	Grounding and CRAG measures	Shows evidence-control behaviour [E5]
audit_level_scores.json	Positive count, percentage, PASS/FAIL	Traces final audit assessment [E5]
run_config.json and run.log	Exact configuration, source paths, execution log	Supports reproduction and audit [E4]

Measured harness run

The production-parser run completed 16 pairs across two reports. It used structure-aware chunks, hybrid retrieval, retrieval_few_shot k=4, v2_evidence_review, Qwen3-8B through Ollama, temperature 0.0 and seed 42. [E4]

It wrote all listed question-level, retrieval, classification, grounding and audit-level artifacts. [E4][E5]

Release gate

The harness contains a production-readiness gate that fails closed unless minimum evidence requirements are satisfied, including independent balanced labels with Insufficient Evidence support and quality thresholds. The current two-report proof set is therefore not production evidence. [E8]

The focused safety test confirms that a one-report, one-positive-label input cannot pass that gate. [E7]

Verification

The offline mock harness was rerun on 28 August 2026 and passed end to end. It is intentionally a mock-based functional check; it does not validate live-model quality. [E7]

Conclusion

The harness delivers the audit trail requested by the assignment. Its limiting factor is evaluation data quality and size, not the absence of structured outputs.

Score-to-evidence trace

Trace level	Saved record	Reviewer can answer
Report	Report ID and source hash	Which source document was scored?
Question	Question ID, gold answer and generated answer	Was this benchmark decision correct?
Retrieval	Returned IDs/scores and gold IDs	Was relevant evidence available to the model?
Generation	Model, prompt, seed, temperature, raw/schema output	Can the result be replayed?
Audit	Positive count, percentage, threshold, PASS/FAIL	How did question outcomes become the final score?

Harness review conclusion

The evaluation harness is the core control that turns an LLM answer into an auditable experiment. It produces machine-readable question-level CSV/JSON files, retrieval and answer metrics, faithfulness measures, audit scores, a run log and frozen configuration. [E4][E5]

The 70% report-level threshold is a reproducible POC scoring rule, not an expert certification rule. The separate release gate remains BLOCKED, which correctly prevents two passing sample reports from being interpreted as production validation. [E5][E10]

Engineering rationale and review use

The harness is a substantive deliverable, not merely a script that prints an accuracy number. It creates the evidence chain needed to challenge an audit score: the report and question are identifiable, the retrieved chunks and model response are retained, the configuration is frozen, and the aggregate score can be reconstructed from individual Yes/No/Insufficient Evidence decisions. [E4][E5][E8]

Its metric design is deliberately multi-layered. Retrieval quality, answer correctness, grounding behaviour, CRAG behaviour and audit score are measured separately because they answer different questions. A high answer score with weak retrieval should prompt investigation, while a high grounding rate without balanced labels should not be mistaken for validated safety. [E5]

The release gate provides an important engineering safeguard: it converts missing benchmark evidence into an explicit BLOCKED result instead of a silent go-live assumption. The current proof therefore has a defensible outcome: the harness works and produces reproducible results, but the available data does not yet meet the evidence threshold for a production-quality claim. [E7][E8]

Repository evidence cited

- [E4] results/production_real_reports/1787846335_de0ded86a7db/run_config.json and run.log: two-report production-parser run configuration and execution log.
- [E5] results/production_real_reports/1787846335_de0ded86a7db/{retrieval_metrics,question_level_llm_metrics,sota_faithfulness_metrics,audit_level_scores}.json: scored metrics and audit decisions.
- [E7] Verification executed 28 August 2026: python -m unittest test_json_parser.py test_adaptive_rag.py test_security_gates.py (6 tests passed) and python test_harness_mock.py (offline mock end-to-end passed).
- [E10] results/final_validation/VALIDATION_SUMMARY.md and its cited run directories: controlled local LLM, ICL, prompt, retrieval, CRAG, chunking and chunk-parameter comparisons completed 28 August 2026.